Topic 12 – Governed, Optimized Workflow

Testing Results Document

Project Name: Governed Enterprise Intelligence Assistant
Project Topic: 12 – Governed, Optimized Workflow
Company Used: GreenTech Solutions Pvt. Ltd. (Fictional)
Testing Status: Partially Implemented – Core Tests Passed

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1. Testing Objective

To verify that the enterprise workflow:

- Processes enterprise data through independent generation tasks.
- Produces an accurate executive report and business insights.
- Generates monitoring and audit information.
- Enforces manager approval validation.
- Blocks saving authorization for rejected, invalid, or missing approval decisions.

2. Test Environment

Sample Input File: `enterprise_raw_data.docx`

Sample Enterprise Data:

- Revenue: ₹500,000
- Expenses: ₹350,000
- Net Profit: ₹150,000
- Net Profit Margin: 30%
- Customer Retention: 85%
- Sales Growth: 12%
- Customer Satisfaction: 4.3/5

3. Approval Validation Test Cases

TC-01: Valid Manager Approval

Input:
`APPROVE`

Expected Result:

- Approval Status: APPROVED
- Manager Decision: APPROVE
- Saving Authorization: AUTHORIZED

Actual Result: APPROVED, AUTHORIZED

Status: PASS

TC-02: Manager Rejection

Input:
`REJECT`

Expected Result:

- Approval Status: REJECTED
- Manager Decision: REJECT
- Saving Authorization: BLOCKED

Actual Result: REJECTED, BLOCKED

Status: PASS

TC-03: Invalid Manager Decision

Input:
`MAYBE`

Expected Result:

- Approval Status: INVALID
- Manager Decision: MAYBE
- Saving Authorization: BLOCKED

Actual Result: INVALID, BLOCKED

Status: PASS

TC-04: Blank Manager Approval

Input: Empty approval response

Expected Result:

- Approval Status: INVALID
- Manager Decision: Blank
- Saving Authorization: BLOCKED

Actual Result: INVALID, BLOCKED

Status: PASS

4. Additional Workflow Verification

Test ID	Test Description	Result
TC-05	Verify independent generation connections	PASS
TC-06	Verify Generate Portal receives four upstream outputs	PASS
TC-07	Verify manager approval connections	PASS
TC-08	Verify executive portal sections	PASS
TC-09	Verify enterprise financial figures	PASS
TC-10	Verify monitoring and audit-log output	PASS
TC-11	Verify pending manager approval status	PASS
TC-12	Verify no false Google Drive save confirmation	PASS

5. Executive Portal Verification

The generated portal was checked for these sections:

1. Executive Summary
2. Financial Performance
3. Customer Performance
4. Operational Performance
5. Business Insights
6. Monitoring and Audit Log
7. Human Approval Status

Result: All seven sections were verified by the user.

6. Data Accuracy Verification

Metric	Source Value	Verification
Revenue	₹500,000	PASS
Expenses	₹350,000	PASS
Net Profit	₹150,000	PASS
Net Profit Margin	30%	PASS
Customer Retention	85%	PASS

7. Governance and Security Verification

- Manager approval is explicitly requested.
- The system does not assume approval.
- REJECTED and INVALID decisions block saving authorization.
- Blank approval is handled as INVALID.
- The report is not falsely described as saved.
- Sharing permissions were restricted using the available Opal settings.

8. Known Limitations

The following requirements remain incomplete in the current Opal configuration:

1. Automatic saving of approved reports to shared company Google Drive is not implemented because the required saving tool was unavailable.
2. Automatic conditional routing based on approval status is not implemented because the required branching functionality was unavailable.
3. Persistent centralized audit-log storage has not been confirmed.
4. Full enterprise RBAC enforcement has not been independently verified.

These limitations must be disclosed during the trainer demonstration.

9. Overall Test Summary

Category	Result
Approval validation	4/4 PASS

Additional workflow checks	8/8 PASS
Executive portal verification	PASS
Data accuracy verification	5/5 PASS
Governance verification	PASS

Overall Assessment: Core workflow generation, report verification, and manager approval validation tests passed. The project is partially implemented because automated Google Drive saving and conditional routing remain outstanding.

10. Conclusion

The Governed Enterprise Intelligence Assistant successfully demonstrates parallel enterprise report generation, structured business reporting, monitoring output, and human-in-the-loop approval validation.

All four manager approval scenarios produced the expected outcomes.

The project demonstrates the core governance principles of monitoring, controlled authorization, and human oversight. Additional integration work is required to complete automatic report storage and conditional workflow execution.

Final Status: CORE TESTS PASSED – PARTIAL IMPLEMENTATION

Prepared for: Trainer Demonstration
Project: Topic 12 – Governed, Optimized Workflow